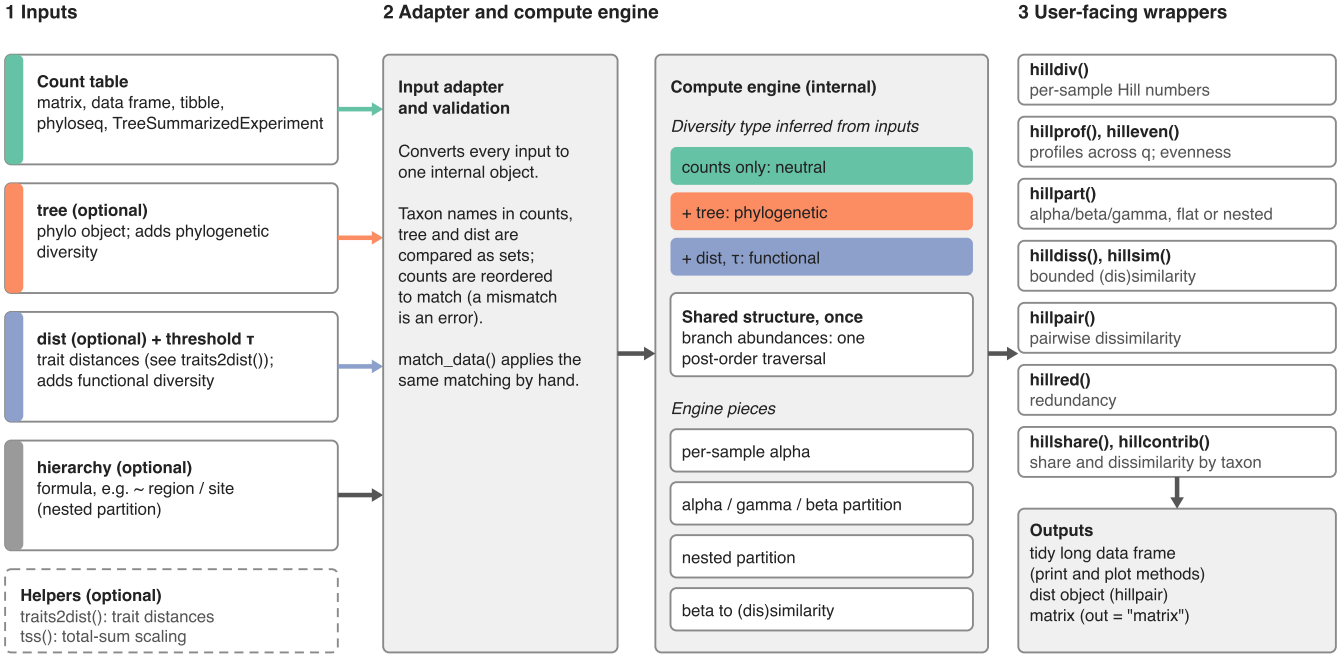

1 Inputs
Count table
matrix, data frame, tibble, phyloseq, TreeSummarizedExperiment
tree (optional)
phylo object; adds phylogenetic diversity
dist (optional) + threshold τ
trait distances (see traits2dist()); adds functional diversity
hierarchy (optional)
formula, e.g. ~ region / site (nested partition)
Helpers (optional)
traits2dist(): trait distances
tss(): total-sum scaling
2 Adapter and compute engine
Input adapter and validation
Converts every input to one internal object.
Taxon names in counts, tree and dist are compared as sets; counts are reordered to match (a mismatch is an error).
match_data() applies the same matching by hand.
Compute engine (internal)
Diversity type inferred from inputs
counts only: neutral
+ tree: phylogenetic
+ dist, τ: functional
Shared structure, once
branch abundances: one post-order traversal
Engine pieces
per-sample alpha
alpha / gamma / beta partition
nested partition
beta to (dis)similarity
3 User-facing wrappers
hilldiv()
per-sample Hill numbers
hillprof(), hilleven()
profiles across q; evenness
hillpart()
alpha/beta/gamma, flat or nested
hilldiss(), hillsim()
bounded (dis)similarity
hillpair()
pairwise dissimilarity
hillred()
redundancy
hillshare(), hillcontrib()
share and dissimilarity by taxon
Outputs
tidy long data frame
(print and plot methods)
dist object (hillpair)
matrix (out = "matrix")